

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE:CSA0706

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

QUESTIONS:4

Total Marks:40

Code of Conduct:

I D.Uday Kiran (192425161) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D.Uday Kiran

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	2	2	1	1	8.5
Q2	2	3	2	1	1	9
Q3	2.5	3	2	1	1	8.5
Q4	2	2.5	2	1	1	9

Faculty Signature

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP. (BL4) (10)

Experiment 1: TCP Three-Way Handshake (SYN, SYN-ACK, ACK)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.flags.syn==1

No.	Time	Source	Destination	Protocol	Length	Info
113	1.843802	172.23.162.119	172.23.22.197	TCP	66	63835 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
522	13.365581	172.23.22.197	4.150.223.107	TCP	66	58301 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
558	13.613034	4.150.223.107	172.23.22.197	TCP	66	443 → 58301 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
733	17.518788	172.25.180.135	172.23.22.197	TCP	66	52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
766	18.440156	172.25.180.135	172.23.22.197	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
880	20.269292	172.25.180.135	172.23.22.197	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 52113 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1

> Frame 113: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF{1B45AFE5-6438-45A3-8C5D-665F59C7C677}, id 0

> Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)

> Internet Protocol Version 4, Src: 172.23.162.119, Dst: 172.23.22.197

> Transmission Control Protocol, Src Port: 63835, Dst Port: 7680, Seq: 0, Len: 0

0000 00 38 fb 90 80 b9 24 d5 e4 6d bc f7 00 00 45 00 - 8 - - - \$ - m - - - E -

0010 00 34 d6 a2 40 00 7f 06 13 b6 ac 17 e2 77 ac 17 - 4 - @ - - - - - w - -

0020 16 c5 f6 3e 1e 00 c4 35 7b 32 00 00 00 00 00 02 - - - - - 5 { 2 - - - - -

0030 ff ff 0a 01 00 00 02 04 05 b4 01 03 03 08 01 01 - - - - -

0040 04 02 - - - - -

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

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> Internet Protocol Version 4, Src: 172.25.180.135, Dst: 172.23.22.197

> Transmission Control Protocol, Src Port: 52113, Dst Port: 7680, Seq: 0, Len: 0

0000 00 38 fb 90 80 b9 24 d5 e4 6d bc f7 00 00 45 00 - 8 - - - \$ - m - - - E -

0010 00 34 d6 a2 40 00 7f 06 13 b6 ac 17 e2 77 ac 17 - 4 - @ - - - - - w - -

0020 16 c5 f6 3e 1e 00 c4 35 7b 32 00 00 00 00 00 02 - - - - - 5 { 2 - - - - -

0030 ff ff 0a 01 00 00 02 04 05 b4 01 03 03 08 01 01 - - - - -

0040 04 02 - - - - -

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> Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)

> Internet Protocol Version 4, Src: 172.25.180.135, Dst: 172.23.22.197

> Transmission Control Protocol, Src Port: 52113, Dst Port: 7680, Seq: 0, Len: 0

0000 00 38 fb 90 80 b9 24 d5 e4 6d bc f7 00 00 45 00 - 8 - - - \$ - m - - - E -

0010 00 34 d6 a2 40 00 7f 06 13 b6 ac 17 e2 77 ac 17 - 4 - @ - - - - - w - -

0020 16 c5 f6 3e 1e 00 c4 35 7b 32 00 00 00 00 00 02 - - - - - 5 { 2 - - - - -

0030 ff ff 0a 01 00 00 02 04 05 b4 01 03 03 08 01 01 - - - - -

0040 04 02 - - - - -

wireshark-Wi-Fi-PF213.pcapng

Packets: 883 · Displayed: 6 (0.7%) · Dropped: 0 (0.0%) · Profile: Default

32°C

Search

11:01

Experiment 2 — DNS Query Using UDP in Wireshark

The screenshot shows the Wireshark interface with a packet capture of a DNS query. The packet list on the left shows packet 276 at time 6.156587, source 192.168.0.239, and destination 255.255.255.255. The packet details pane on the right shows the following structure:

- Header Checksum: 0x7919 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 192.168.0.239
- Destination Address: 255.255.255.255
- User Datagram Protocol, Src Port: 53, Dst Port: 53
 - Source Port: 53
 - Destination Port: 53
 - Length: 41
 - Checksum: 0x3317 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 88]
 - [Timestamps]
 - UDP payload (33 bytes)
 - Domain Name System (query)
 - 0000 ff ff ff ff ff ff ff ff e0 46 ee 38 96 d0 08 00 45 00F 8...E
 - 0010 00 3d 00 00 40 00 40 11 79 19 c0 a8 0e ff ff@.y.....
 - 0020 ff ff 00 35 00 35 00 29 33 17 63 8e 01 00 00 015-5) 3 c.....
 - 0030 00 00 00 00 00 04 74 69 6d 05 06 67 6f 6f 67t ime goog
 - 0040 6c 65 03 63 6f 6d 00 00 1c 00 01 00 85 09 00 00 ..e.com.....
 - 0050 00 00 00 00 00 00 00 00 00 00 00 03 00 01 4c 00L.....
 - 0060 00 00 2a 00 00 00 05 c0 02 00 78 47 12 00 00 80xg.....
 - 0070 27 00 32 00 00 00 6d 3f 81 16 4c 00 00 00 64 43 ..2...m? ..L...dC
 - 0080 81 16 4c 00 00 00 ff ff ff ff 00 00 00 00 00 00L.....
 - 0090 00 00 cd 02 f1 02 00 00 00 00 00 00 00 ac 09A t...ls
 - 00a0 01 00 00 00 7c a7 88 41 74 00 80 cc 9c 6c 73 e2#5...m...r,
 - 00b0 48 ad 9a 96 1e 23 24 d5 e4 6d bc f7 60 08 10 2cE-(-@...9...;
 - 00c0 7d 00 00 20 00 00 00 00 aa aa 03 00 00 00 08 005...GN X...
 - 00d0 45 00 00 28 af 85 40 00 00 06 39 e5 ac 17 18 3bP.....n...
 - 00e0 17 c9 35 8a f0 12 01 bb 85 92 47 4e 09 58 2d 99?.....
 - 00f0 50 10 00 ff a7 90 00 00 e3 e5 6e d1 8b 0b 60 a5P.....
 - 0100 01 3f 00 00 05 20 00 00 00 00 00 00 00 01 002.....
 - 0110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00P.....

The packet bytes pane at the bottom shows the raw hex and ASCII data for the packet.

The screenshot shows the Wireshark interface with a packet capture of a DNS query. The packet list on the left shows packet 275 at time 6.286529, source 192.178.211.188, and destination 172.23.22.197. The packet details pane on the right shows the following structure:

- Frame 275: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1845AFE5-6438-45A3-8C5D-66F59C7C677}, id 0
- Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: IntelCor_90:88:b9 (80:38:fb:90:88:b9)
- Internet Protocol Version 4, Src: 192.178.211.188, Dst: 172.23.22.197
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0xb0 (DSCP: CS4, ECN: Not-ECT)
 - Total Length: 52
 - Identification: 0xb080 (48648)
 - Flags: 0xb0
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 122
 - Protocol: TCP (6)
 - Header Checksum: 0x2af0 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.178.211.188
 - 0000 80 38 fb 90 88 b9 24 d5 e4 6d bc f7 08 00 45 808...\$...m...E
 - 0010 00 34 be 08 00 00 7a 06 2a f0 c0 b2 d3 bc ac 174...2...*.....
 - 0020 16 c5 14 6c d3 e1 7e aa 67 ab fe 82 23 71 80 101...g...#q...
 - 0030 04 17 e9 db 00 00 01 01 05 0a fe 82 23 70 fe 82#p...
 - 0040 23 71#q

The packet bytes pane at the bottom shows the raw hex and ASCII data for the packet.

Experiment 3 — TCP Retransmission vs UDP Packet Loss

The image displays two screenshots of the Wireshark network protocol analyzer. The top screenshot shows a TCP retransmission event. The bottom screenshot shows a UDP packet loss event.

Top Screenshot: TCP Retransmission

The packet list shows two TCP packets from 172.25.180.135 to 172.23.22.197. The first packet (No. 766) is a SYN packet (Seq=52113, Win=0) that was retransmitted. The second packet (No. 800) is a SYN packet (Seq=52113, Win=0) that was also retransmitted. The packet details pane shows the TCP header for the first packet, indicating it is a SYN packet with Seq=52113, Win=0, and a length of 66 bytes.

Bottom Screenshot: UDP Packet Loss

The packet list shows a single UDP packet (No. 276) from 192.168.0.239 to 255.255.255.255. The packet details pane shows the UDP header for the first packet, indicating it is a UDP packet with Src Port=53, Dst Port=53, and a length of 593 bytes. The packet details pane also shows the User Datagram Protocol (UDP) header, indicating it is a UDP packet with Src Port=53, Dst Port=53, and a length of 53 bytes.

Experiment 4 — DNS Resolution in Wireshark

The image shows a Wireshark packet capture of a DNS query. The packet list on the left shows a single packet at time 276.6150587, source 192.168.0.239, destination 255.255.255.255, protocol DNS, length 593. The packet details pane shows the following structure:

- Frame 276: 593 bytes on wire (4744 bits), 593 bytes captured (4744 bits) on interface \Device\NPF_{1B45AFE5-6438-45A3-8C5D-665F59C7C677}, id 0
- Ethernet II, Src: Netgear_38:96:d0 (e0:46:ee:38:96:d0), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Internet Protocol Version 4, Src: 192.168.0.239, Dst: 255.255.255.255
- DNS
- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 61
- Identification: 0x0000 (0)
- Flags: 0x40, Don't fragment
- ... 0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: UDP (17)
- Header checksum: 0x7919 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 192.168.0.239

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII part shows the query: "Standard query 0x638e AAAA time.google.com".

At the bottom of the Wireshark window, the status bar shows: Packets: 883 · Displayed: 1 (0.1%) · Dropped: 0 (0.0%)